

ER Diagram of Database

Architecture and Features

The project is organized with Flask handling the backend logic and routing. Routes(Controllers) are defined in Python files within a `routes` directory, managing the application's various endpoints and interactions.

The **templates** used for rendering HTML are stored in the `**templates**` directory, utilizing Jinja2 for dynamic content. This structure allows HTML files to be cleanly separated from backend logic.

Static assets like CSS, JavaScript, and images are placed in the `**/static**` directory, ensuring they are accessible for styling and functionality.

Features implemented in the application include:

- **Admin Login and User Login**
 - Admin Login: Admins log in using a dedicated form.
 - Separate Signup: Sponsors and influencers have separate signup forms, while a common login form is used for all users.
 - Technology: Flask (backend logic), HTML (forms), Flask-SQLAlchemy (user model for differentiating between user types).
- **Admin Dashboard**
 - Flagged Users/Campaigns: Displays information on flagged users and campaigns with options to flag or unflag.
 - Statistics: Shows data on active users, campaigns, ad requests, and flagged items.
 - Technology: Flask (backend), HTML and Bootstrap (dashboard UI), Flask-SQLAlchemy (data retrieval), matplotlib (stats).
- **Campaign Management**
 - Create Campaigns: Allows sponsors to create new campaigns.
 - Update Campaigns: Sponsors can update campaign details like start date, end date, and budget.
 - Delete Campaigns: Sponsors can delete campaigns.
 - Technology: Flask (backend), HTML (forms), Flask-SQLAlchemy (database interactions).
- **Ad Request Management**
 - Create Ad Requests: Sponsors can create ad requests based on campaign goals.
 - Edit Ad Requests: Modify existing ad requests, including details like influencer_id and payment amount. i.e. (Negotiate Feature allows update of ad request)
 - Delete Ad Requests: Remove ad requests as needed.
 - Technology: Flask (backend), HTML (forms), Flask-SQLAlchemy (database operations).
- **Search for Influencers and Public Campaigns**
 - Sponsor Search: Sponsors search for influencers based on criteria such as niche, reach, and followers.
 - Influencer Search: Influencers is viewed with the campaigns of his niche.
 - Technology: Flask (search functionality), HTML (search forms), Flask-SQLAlchemy (database queries).
- **Action on Ad Requests**
 - View Ad Requests: Influencers can view all ad requests from different campaigns.
 - Accept/Reject Requests: Influencers can accept or reject specific ad requests.
 - Negotiate Payment: Ability to negotiate payment amounts for ad requests.
 - Technology: Flask (backend), HTML (interaction forms), Flask-SQLAlchemy (data updates).
- **Statistics Visualization**
 - Campaign Performance: Matplotlib generates charts and graphs to visualize campaign performance metrics.
 - User Statistics: Various user statistics are visualized through comparisons and distributions.
 - Technology: Matplotlib (charting and graphing).

Video

Link:

https://drive.google.com/file/d/18iBzrww7pKGFNaX1Fz1pbYzJamaMlach/view?usp=drive_link